PREDICTING WINE QUALITY ANALYSIS

Sikawura Kobina Ebo Arkorful

April 2026

TABLE OF CONTENT

Problem Statement

The Data

Exploratory Data Analysis 1: (Distribution)

Exploratory Data Analysis 2: (Correlations)

Data Preprocessing

Modeling Approach

Model Performance

Feature Importance

Conclusion & Recommendations

PROBLEM STATEMENT

Wine quality has long been assessed by human tasters, a method that can be subjective, costly, and difficult to apply at scale for larger producers.

My mission is to uncover the chemical 'fingerprint' indicative of high-quality wines

By accurately predicting quality early in production, a winery can:

i. Make real-time adjustments to fermentation processes, protecting batches that are at risk.

ii. Implement strategic pricing for batches grounded in objective chemical analysis.

iii. Uphold brand consistency across both red and white wine varieties, enhancing customer trust and loyalty.

DATA OVERVIEW

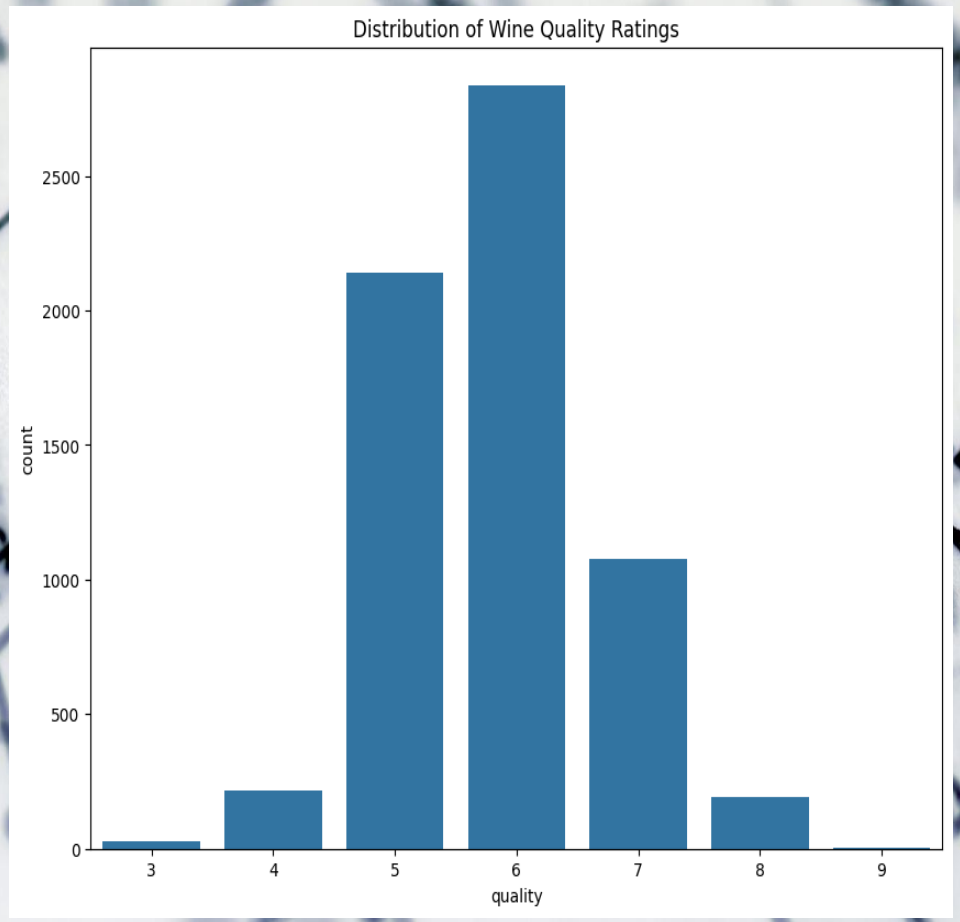
To have a broader scope, I combined the Red & White datasets, handling 6,497 rows) and added wine type feature, assigned;

- 0 for Red and
- 1 for White wine.

A quick overview indicated a well-structured dataset without missing values.

- Features include 11 chemical properties (eg. Alcohol, Acidity, Chlorides, and pH)

EXPLORATORY DATA ANALYSIS (DISTRIBUTION)



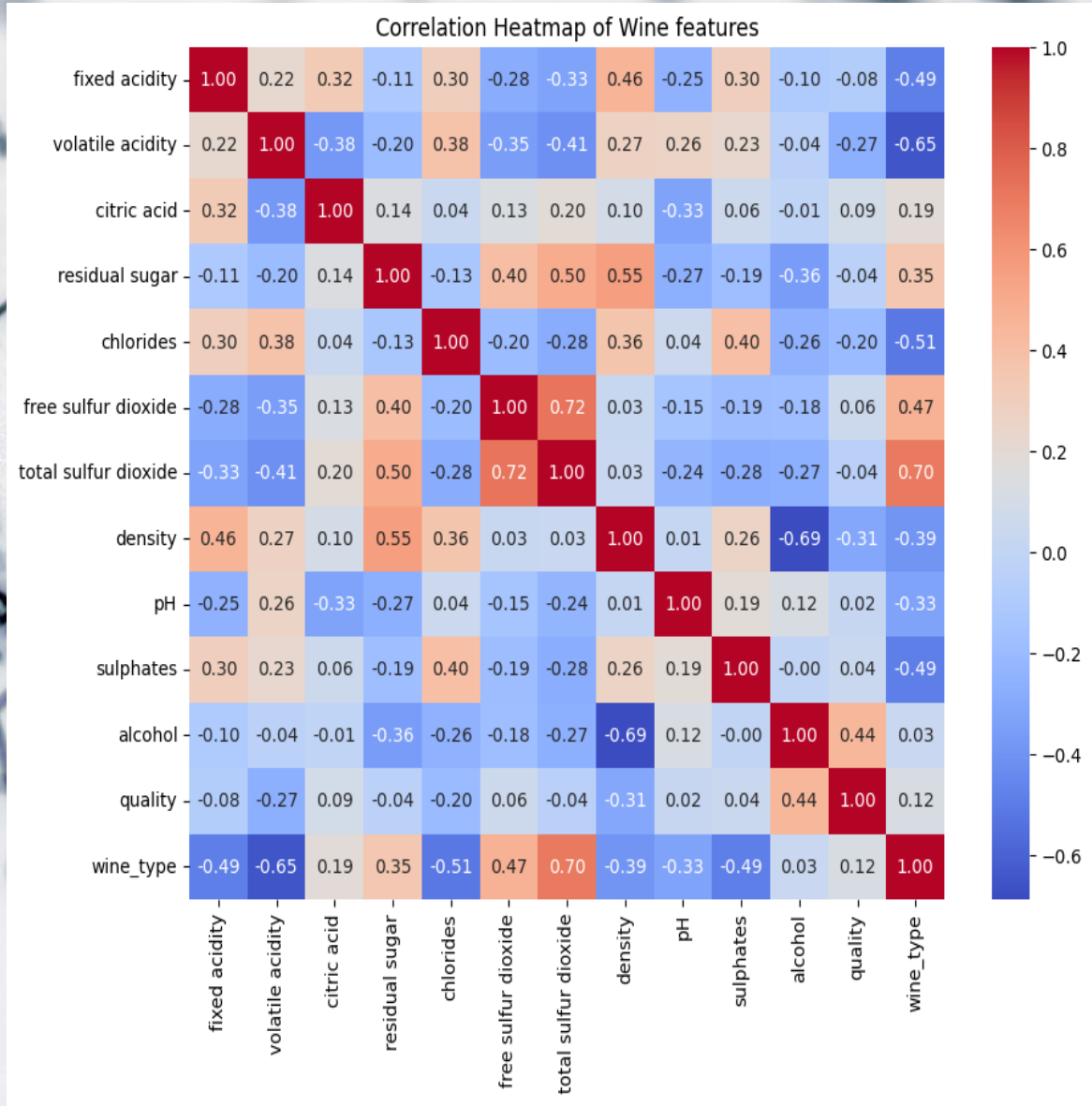
Averagely: Most wines in the dataset are rated 5 or 6, creating a "Normal Distribution" centered around average quality.

The Imbalance Challenge:

80.3% Standard Wines

19.7% Good Wines

Strategic Impact: This imbalance requires specialized techniques (like SMOTE) to ensure the model doesn't simply "ignore" the premium wines to achieve high accuracy.



EXPLORATORY DATA ANALYSIS (CORRELATIONS)

- **Heatmap Analysis:** Identified key physicochemical properties that influence sensory ratings.
- **Primary Quality Drivers: Alcohol c**ontent shows the strongest positive correlation with quality (r = 0.44), suggesting higher ripeness and body are favored.
- **The "Quality Killer": Volatile acidity** shows a strong negative correlation. Higher levels (associated with a vinegar-line taste) significantly detract from quality ratings.
- **Feature Significance:** These results confirm that the features are both interpretable and highly relevant for the upcoming classification models

DATA PREPROCESSING

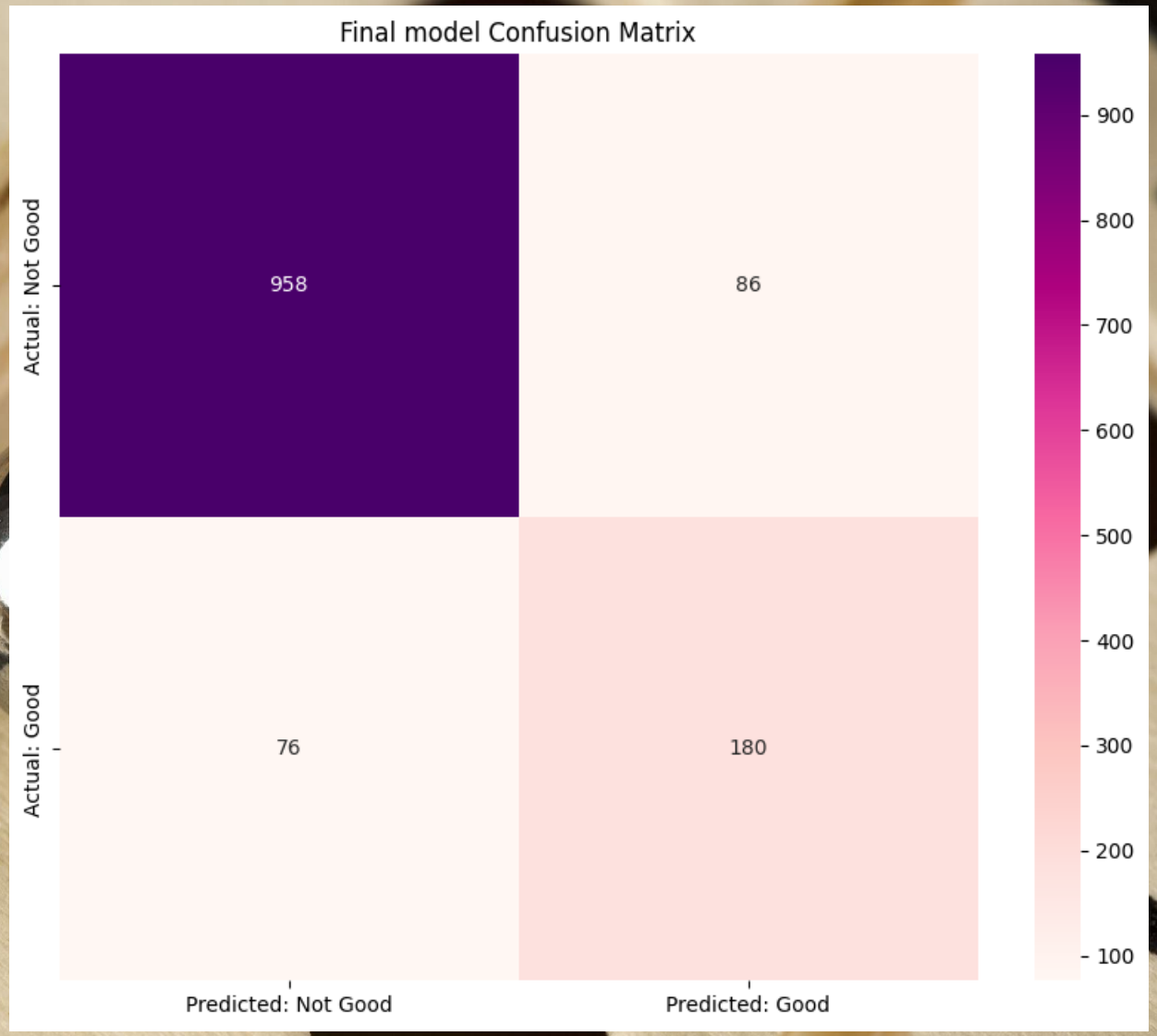
- **Feature Scaling:** Used StandardScaler to normalize features like density and sulfur dioxide, ensuring the model isn't biased by different units of measurement.
- **Preventing Leakage:** Scaler was fit only on the training data to ensure the model had no "peek" at the test set distribution.
- **Class Imbalance (SMOTE):** Addressed the 80/20 split between "Standard" and "Good" wines.
  Why? Without SMOTE, the model would over-prioritize the majority class (Standard wines) and fail to learn what makes a wine "Premium."
- **Evaluation Metric:** Shifted focus from Accuracy to F1-Score. Because "Good" wines are rare, the F1-Score provides a more honest view of how well we are catching the premium products.

MODELING STRATEGY

Logistic Regression: Used as a baseline for linear relationships.

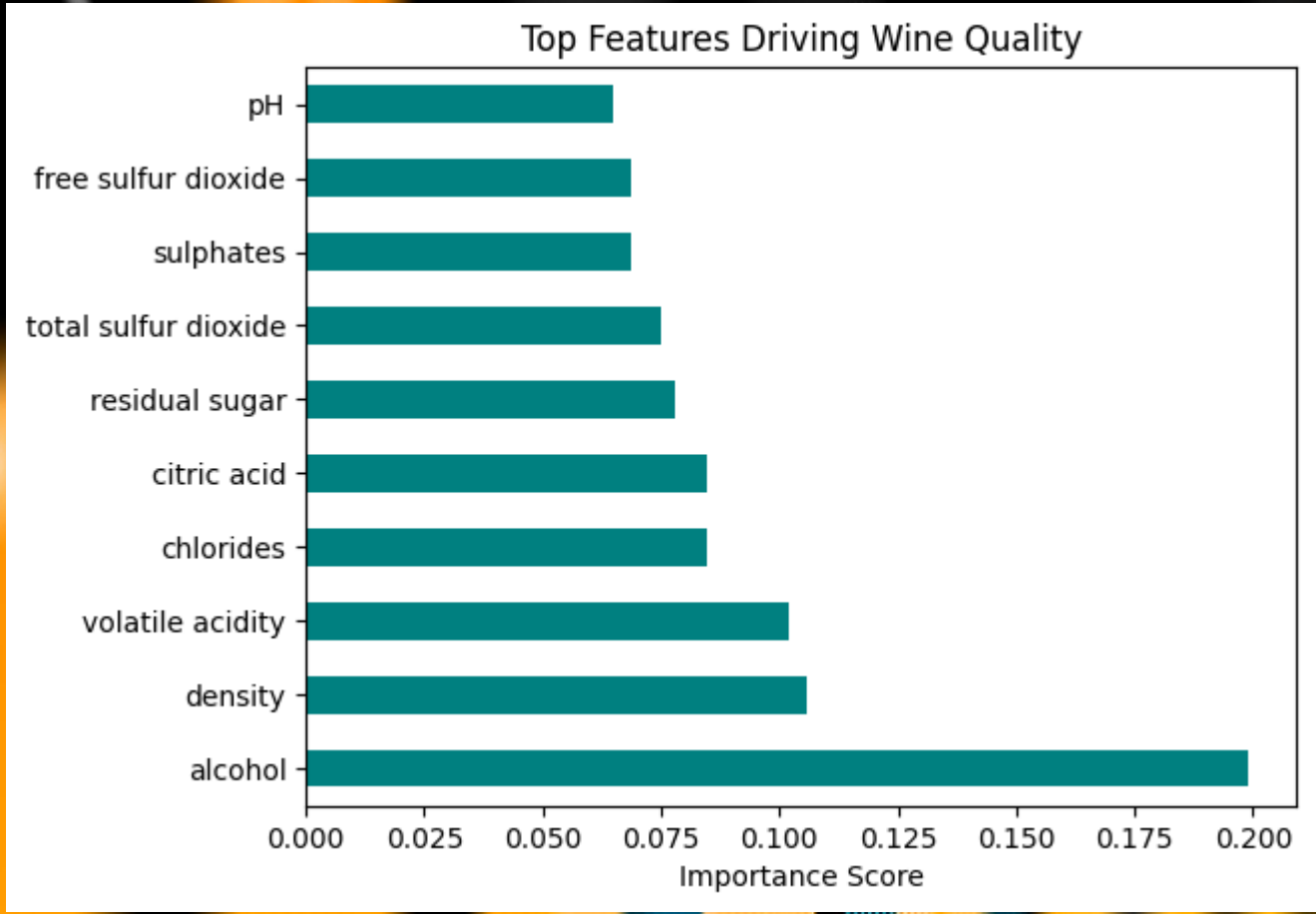
Random Forest: Selected for its ability to handle non-linear interactions between chemical properties.

Optimized using **GridSearchCV** to maximize F1-Score.



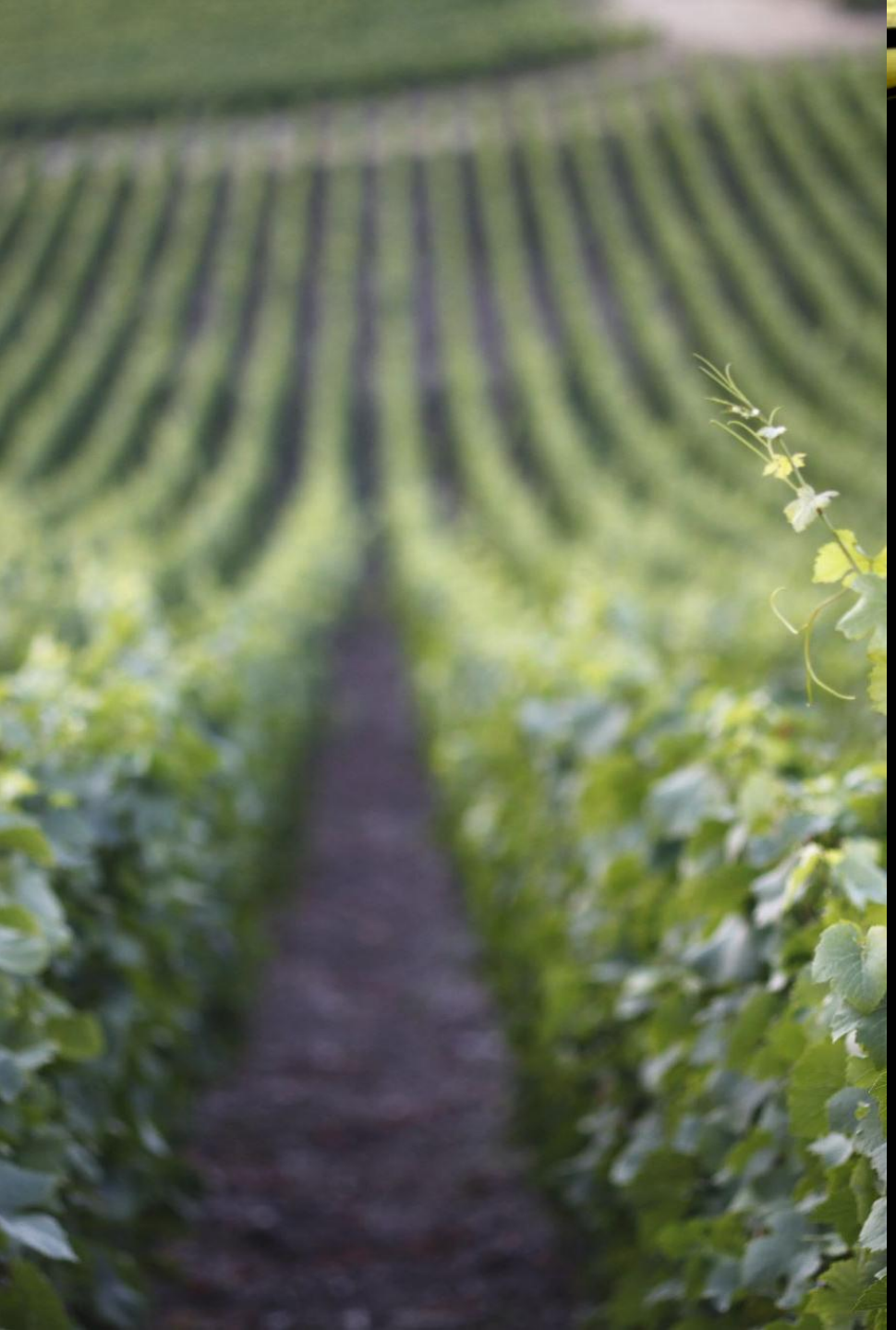
MODEL PERFORMANCE

- **Final Accuracy: 88%**
- This is a strong improvement over the "baseline" accuracy (80%) of simply guessing the most common class.
- **The "Good" Wine Success (F1-Score: 0.69):**
- **Precision (0.68):** When the model flags a wine as "Premium," it is correct 68% of the time.
- **Recall (0.70):** The model successfully catches 70% of all high-quality wines in the dataset.
- **The Power of Tuning:**
- Through **GridSearchCV**, hyperparameters were optimized to maximize the F1-Score, ensuring a balance between catching quality wines and avoiding "False Alarms."



KEY DRIVERS OF QUALITY

- **Alcohol as the Primary Indicator:** Consistent with our EDA, Alcohol remains the strongest predictor of high-quality ratings.
- **The Impact of Volatile Acidity:** As the second or third most important feature, this confirms that controlling "vinegar-like" acidity is vital for premium wine production.
- **Density & Chlorides:** These features also rank high, suggesting that the "mouthfeel" (body and saltiness) plays a significant role in sensory evaluation.
- **Model Interpretability:** The alignment between our initial correlations (Heatmap) and the Random Forest's feature importance provides high confidence in the model's logic.



CONCLUSION & RECOMMENDATIONS

Conclusion

- **Predictive Success:** We successfully developed a model that achieves 88% accuracy and, more importantly, identifies 70% of premium wines despite a significant class imbalance.
- **Technical Validation:** The use of SMOTE and Random Forest proved to be the correct strategy for handling the complexities and non-linear relationships in sensory wine data.

Future Recommendations:

- **Automated Quality Sorting:** This model can be integrated into the bottling line as an automated "Early Warning System" to flag batches that may need further manual testing or price adjustments.
- **Production Guidance:** Winemakers should prioritize the management of Alcohol and Volatile Acidity, as the model identifies these as the primary gatekeepers of a "Premium" rating.
- **Expanding the Scope:** Future iterations could include geographic data (region of origin) or vintage year to further refine the accuracy of quality predictions.

”Premium wine never ceases to satisfy the soul, especially when it’s squeezed, blended, processed, and stored for a period.”

THANK YOU!!